

Hermit Interpolation

Say, x_0 and $f(x_0)$ is given. Now we need another term which is $f'(x_0)$.

$n=2$

x	$f(x)$	$f'(x)$
$x_0 \dots 1$	10	2
$x_1 \dots 5$	20	4
$x_2 \dots 10$	30	6

3 conditions/
nodes
($n+1$)

3 conditions/
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($n+1$)

$\Rightarrow$ total $(3+3) = 6$ conditions

if n is degree,
we need $(2n+1)$
conditions/nodes

Here, $n=2$,

so, total polynomial
degrees $= 2 \times 2 + 1 = 5$

Hermit Expression

$$P_{2n+1} = h_0(x) f(x_0) + h_1(x) f(x_1) + \dots + \hat{h}_0 f'(x_0) + \hat{h}_1 f'(x_1) + \dots + \hat{h}_n f'(x_n)$$

Here,

$$\Rightarrow h_k(x) = \{ 1 - 2(x - x_k) l'_k(x_k) \} l_k(x)^2$$

$$\Rightarrow \hat{h}_k = (x - x_k) l_k(x)^2$$

Ex-1 $f(x) = \sin(x)$

$f'(x) = \cos(x)$

x	$f(x)$	$f'(x)$
$x_0 \dots 0$	0	1
$x_1 \dots \pi/2$	1	0

So,

$$P_{2 \times 1+1} \Rightarrow P_3(x) = h_0 f(x_0) + h_1 f(x_1) + \hat{h}_0(x) f'(x_0) + \hat{h}_1(x) f'(x_1)$$

$$= \underline{h_1(x)} f(x_1) + \underline{\hat{h}_0(x)} f'(x_0) \dots \textcircled{i}$$

Formula: $h_k(x) = \{ 1 - 2(x - x_k) \underline{l'_k(x_k)} \} * (\underline{l_k(x)})^2$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{\pi/2 - 0} = \frac{2x}{\pi}$$

$$l'_1(x) = 2/\pi$$

$$\hat{h}_k(x) = (x - x_k) (l_k(x))^2$$

$\Rightarrow$ we need $\hat{h}_0(x)$.

$$\hat{h}_0(x) = (x - 0) \left(1 - \frac{2x}{\pi}\right)^2$$

$$= x \left(1 - \frac{2x}{\pi}\right)^2$$

we need

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - \frac{\pi}{2}}{0 - \frac{\pi}{2}} = 1 - \frac{2x}{\pi}$$

From (i) $\Rightarrow$

$$P_3(x) = h_1 f(x_1) + \hat{h}_0(x) f'(x_0)$$

$$= \left\{ 1 - 2\left(x - \frac{\pi}{2}\right) * \frac{2}{\pi} \right\} * \left(\frac{2}{\pi}\right)^2 * 1 + \left(x - \left(1 - \frac{2x}{\pi}\right)^2\right) * 1$$

this is the desired expression of the table!

Ex-2

	x	$f(x)$	$f'(x)$
$n=2$	$x_0 \dots 0$	1	0
	$x_1 \dots 1$	0	1
	$x_2 \dots 2$	2	0

$$\begin{aligned} \therefore P_{2-2+1} \Rightarrow P_5(x) &= h_0(x) f(x_0) + \cancel{h_1(x) f(x_1)} + h_2(x) f(x_2) \\ &\quad + \cancel{\hat{h}_0(x) f'(x_0)} + \hat{h}_1(x) f'(x_1) + \cancel{\hat{h}_2(x) f'(x_2)} \\ &= h_0(x) f(x_0) + h_2(x) f(x_2) + \hat{h}_1(x) f'(x_1) \end{aligned}$$

Formulas:

i) $h_k(x) = \{ 1 - 2(x - x_k) l'_k(x_k) \} * \{ l_k(x) \}^2$

ii) $\hat{h}_k(x) = (x - x_k) (l_k(x))^{n-1}$

Let's calculate

need $l_0(x)$ for $h_0(x) \Rightarrow l_0(x) = \frac{x - x_1}{x_0 - x_1} * \frac{x - x_2}{x_0 - x_2}$

$$= \frac{x - 1}{-1} * \frac{x - 2}{0 - 2}$$

$$= \frac{1}{3} (x - 1)(x - 2)$$

$h_0(x) = \{ 1 - 2(x - 0) * \frac{1}{3}(2x - 3) \} * \{ \frac{1}{3}(x - 1)(x - 2) \}^2$

For $h_2(x) \Rightarrow$ Find $l_2(x)$ and $l'_2(x)$

Similarly we get $\frac{x(x-1)}{2}$ and $\frac{2x-1}{2}$

$\therefore h_2(x) = \{ 1 - 2(x - 2) * \frac{2x-1}{2} \} * \left(\frac{x(x-1)}{2} \right)^2$

Now need $\hat{h}_1(x)$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} \times \frac{x-x_2}{x_1-x_2} \\ = \frac{x-0}{1-0} \times \frac{x-2}{1-2} = \frac{x(x-2)}{-1}$$

$$\therefore \hat{h}_1(x) = (x-x_1) (l_1(x))^2 \\ = (x-1) \left(\frac{x(x-2)}{-1} \right)^2$$

After plugging all values

$$P_5(x) = \underbrace{h_0(x)}_{1-2x} \underbrace{f(x_0)}_{\frac{1}{3}(2x-3)} + \underbrace{h_2(x)}_{1-2(x-2) \times \frac{2x-1}{2}} \underbrace{f(x_2)}_{\frac{x(x-1)}{2}} + \hat{h}_1(x) f'(x_1)$$

$$= \left\{ 1-2x \times \frac{1}{3}(2x-3) \right\} \times \left\{ \frac{(x-1)(x-2)}{3} \right\} \times 1 + \left\{ 1-2(x-2) \times \frac{2x-1}{2} \right\} \times \left(\frac{x(x-1)}{2} \right) \times 2$$

$$+ (x-1) \left(\frac{x(x-2)}{-1} \right)^2 \cdot 1$$

Aus

this is
the expression